



Bilateral tax competition in double tax treaties between developed and developing countries

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Accepted: 28 January 2025
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Abstract

This paper investigates tax competition in withholding tax rates signed under double tax treaties between developed and developing countries. We document a decline in average withholding tax rates since 1990 and attribute this trend to tax competition. As opposed to traditional tax competition, where every foreign investor benefits from lower tax rates, tax treaties allow countries to compete bilaterally, cutting tax rates for investors from a specific country and leaving taxes for everyone else unaffected. We test our predictions empirically for the four different types of withholding tax rates on passive income in more than 900 double tax treaties. We find strong support for a positive relationship in withholding tax rates in tax treaties signed by developing countries in relation to the same origin country of investment, supporting the notion of tax competition.

Keywords Tax competition · International taxation · Double taxation treaties · Tax treaty formation · Withholding tax rates

JEL Classification F50 · F53 · F68 · H29 · H39

1 Introduction

This paper investigates tax competition in withholding tax rates signed under double tax treaties (DTTs) between developed and developing countries. DTTs remain an important policy tool to promote cross-border economic activity by preventing international double taxation. Traditionally, DTTs provide for a bilateral framework to

Kunka Petkova work has been partly completed before the author joined the German Federal Ministry of Finance. The views expressed in this paper are those of the author only and do not involve the responsibility of the German Federal Ministry of Finance. Andrzej Stasio work has been partly completed before the author joined the European Commission Joint Research Centre. The views expressed in this paper are those of the author only and do not involve the responsibility of the European Commission.

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allocate taxing rights between the two contracting states and set a binding maximum on source (i.e. destination) country withholding tax rates on passive income. However, over the years, DTTs have come to pursue additional goals such as providing legal certainty, preventing tax discrimination in the state of investment and exchange of information for tax matters.

We focus on the role of source country withholding taxes and document their development between 1990 and 2016 for the universe of tax treaties signed by developed countries. While the average withholding tax rate under DTTs signed between two developed countries remains fairly constant during this time period, we observe a significant decline in the average withholding tax rate under DTTs signed between developed and developing countries. We argue that this trend is consistent with the notion of tax competition and investigate our hypothesis empirically.

We leverage on the time dimension of treaty formation and analyse our hypothesis for the four distinct types of withholding tax rates: on portfolio dividends, participation dividends, interest and royalties. In particular, we resort to a space-time regression model and regress the observed withholding tax rate in the year of treaty conclusion on the lagged average withholding tax rate from *past* tax treaties signed by other developing countries in relation to the same origin country of investment.

We find strong support for a positive relationship in withholding tax rates, consistent with the notion of tax competition for mobile capital. Withholding tax rates concluded between developed and developing countries decrease in the average withholding tax rate previously negotiated by other developing countries in relation to the same developed country. Depending on the type of the withholding tax rate, our results suggest that the observed withholding tax rate will be between 0.31 pp. and 0.48 pp. lower for a 1 pp. lower average withholding tax rate from peer countries' past treaties. Moreover, we find a significantly stronger reaction among *open* economies, with elasticities ranging from 0.53 to 0.78. This effect is particularly pronounced for the withholding tax rates on participation dividends, which—in the baseline model specification—is statistically not significant when not distinguishing between *open* and *closed* economies, but is comparable to the other types of withholding tax rates among *open* economies.

We corroborate our results across a wide range of model specifications and robustness tests. We use tax treaty fixed effects to control for unobservable treaty characteristics and continue to find statistically significant results. Next, we reject the hypothesis that our results are driven by yardstick competition instead of tax competition by distinguishing between open and closed economies. We confirm that our results are robust to sample selection bias and possible endogeneity of the weighted average withholding tax rate. We control for indirect tax rates and verify that treaty shopping opportunities do not alter our findings. Finally, we validate our baseline results using alternative definitions of the weighting matrix of peer countries' withholding tax rates.

In extension of our main analysis, we find that the tax competition effect increases for countries in the same geographical regions. Furthermore, more recent tax treaties signed by other developing countries matter more than the older ones, in particular for withholding tax rates on participation dividends. In contrast, economic similarity tends to dampen the reported tax competition effects.

Our research bears important policy implications. On October 8, 2021, 136 member jurisdictions of the OECD/G20 Inclusive Framework on Base Erosion and Profit Shifting (IF) agreed on a two-pillar solution to address the tax challenges arising from the digitalisation of the economy.¹ For companies in scope, Pillar Two of this agreement introduces a global effective minimum corporate tax rate of 15%, effectively putting a floor on tax competition on corporate income tax rates and levelling the playing field by providing a set of rules on how the tax base will be defined. As a consequence, countries that wish to continue using tax policy to compete for inward investment may shift their attention to other areas of business taxation, including labour taxes, indirect taxes and withholding taxes at source. Indeed, our results suggest a considerable scope for DTTs as a tool of tax competition among developing countries.

From a policy perspective, tax competition over withholding tax rates signed under DTTs is fundamentally different from the more traditional form of tax competition over corporate income tax rates. As opposed to traditional tax competition, where every foreign investor would benefit from lower tax rates, DTTs allow countries to compete for investors from a single, specific origin country, while leaving taxes for everyone else unaffected. In this way, tax competition becomes more targeted or—as we would like to phrase it—*bilateral*.

Our paper contributes to various strands of research. First, we contribute to the literature on tax competition. In small open economy models of tax competition, the strategic interactions between the individual countries lead to a race-to-the-bottom and the tendency to abandon capital taxes (Zodrow and Mieszkowski 1986; Wilson 1986). Recent literature documents empirical findings in support of international tax competition and strategic interactions between corporate tax rates.

Egger et al. (2007) find a positive reaction function for both corporate and personal income tax rates in a panel of 30 OECD countries between 1985 and 2005. Devereux et al. (2008) develop a model, in which firms choose where to locate their capital in response to effective marginal tax rates and simultaneously allocate their profit following the differences in statutory tax rate. The authors find evidence of strategic competition over both measures utilising data from 21 OECD countries between 1982 and 1999.

Analysing different competitive behaviours, Redoano (2014) finds evidence of fiscal interdependencies consistent with the literature on tax and yardstick competition. Noteworthy, the regression results suggest that for corporate taxes, European countries follow large countries in order to attract capital. Heinemann et al. (2010) focus not only on the tax-rate-cutting decisions, but also on how these are affected by the tax rates in neighbouring countries. The authors consider 32 European countries, and find that a country is more likely to decrease its corporate tax rate if its own rate is high, while the one of its neighbours is low.

More recently, using a global sample of countries for the period 1995–2014, Azémar et al. (2020) show that strong growth performance of neighbouring countries

¹ See <https://www.oecd.org/tax/beps/brochure-two-pillar-solution-to-address-the-tax-challenges-arising-from-the-digitalisation-of-the-economy-october-2021.pdf>.

is associated with lower corporate income tax rates in the observed country. The effect is especially present in large developed countries, which are open to capital flows. Nonetheless, some developing regions appear to compete indirectly by reducing their corporate income tax rates in response to foreign economic growth.

Synthesising these results, Heimberger (2021) applies meta-analysis and meta-regression methods to a novel dataset consisting of 33 primary studies. Overall, he finds empirical evidence indicating the presence of corporate tax competition and no consistent evidence for publication selection bias. Corporate tax competition is not only statistically significant, but also relevant from an economic point of view.

We extend the existing literature in two directions. First, we show that corporate tax competition extends beyond corporate income tax.² Traditional tax competition literature postulates the idea that countries underbid each other's corporate income tax rates in order to attract mobile capital. This is a rather crude measure as it also reduces corporate income tax rates for investors that are already subject to higher tax rates in the source country, thus giving them a windfall profit.

In contrast, this paper shows empirically that countries (also) engage in a more subtle form of tax competition. Destination countries employ double tax treaties to bilaterally reduce withholding tax rates for foreign investors from a specific country of origin, leaving tax rates with respect to all other countries unaffected—a policy which we label *bilateral* tax competition.

Second, where previous literature focuses primarily on tax competition between developed countries, we provide strong support for tax competition among developing countries on mobile capital from developed countries. We add to the currently limited body of literature on tax competition in developing countries, most notably Azémar et al. (2020) and Crivelli et al. (2015), who estimate government response to foreign changes in corporate income tax rates among OECD and non-OECD countries over the period 1980–2013.³

We also contribute to the literature on tax treaty formation and more specifically, on tax treaty bargaining. Originally, double tax treaties have been serving as an important policy tool to promote international economic activity by preventing international double taxation. However, over the years, DTTs have come to pursue additional goals such as providing legal certainty, preventing tax discrimination in the state of investment and exchange of information for tax matters.

Against these different goals, Ligthart et al. (2011) study empirically the determinants of DTT formation for a large sample of more than 17,000 country-pairs covering the 1950–2006 period. Using a gravity framework, they conclude that countries sign DTTs primarily to reduce international double taxation and, to a lesser extent, to provide a legal instrument for the exchange of information in tax matters. In support of this finding, Davies (2003a, b), argues that the main role of DTTs lies in

² In this regard, our work complements Klemm and Van Parys (2012), who find evidence of tax competition using tax incentives for a sample of 40 Latin American, Caribbean and African countries over the period 1985–2004.

³ In a recent contribution, Heckemeyer et al. (2024) document the average withholding tax rates concluded by developing countries with developed and other developing countries.

the harmonisation and the lowering of withholding tax rates on international capital income.

Chisik and Davies (2004) discuss how the distributional implications of DTTs can affect withholding taxes in the framework of tax treaty bargaining. They predict that more asymmetric countries will conclude treaties with higher withholding tax rates. This theory is then tested using data on U.S. and OECD bilateral tax treaties and the results broadly support their predictions. Rixen and Schwarz (2009) confirm this finding using data on German tax treaties. Nevertheless, these studies view tax treaty negotiations as an exclusively bilateral outcome.

Two noticeable exception in this context has been presented by Barthel and Neumayer (2012) and Arel-Bundock (2017). Barthel and Neumayer (2012) analyse spatial diffusion in tax treaty formation. They show that the probability of two countries entering into a DTT increases with specific *source* and *target contagion*. In other words, the likelihood of two countries signing a tax treaty raises with the origin country's peers having a tax treaty with a given target (destination) country and, conversely, target (destination) country's peers having a tax treaty with the specific origin country of investment. However, their analysis is limited to countries' decision to enter into a DTT and does not extend to the negotiated treaty conditions.

Arel-Bundock (2017) analyses DTTs in the context of the international tax system.⁴ Seen from a network perspective, DTTs create opportunities for treaty shopping, reducing the potency of fiscal policy and putting pressure on governments to change their domestic tax laws. To systematically assess the effect of bilateralism on states' fiscal autonomy, the relationship between bilaterally applicable withholding tax rates and indirect (i.e. treaty shopping) rates is tested empirically in a cross-sectional sample of 164 countries, with the results pointing towards a positive association between direct and indirect tax rates.

We complement the previous literature by studying empirically the determinants of withholding tax rates concluded in DTTs between developed and developing countries. We show that withholding tax rates are a product of tax competition and depend on withholding tax rates signed under DTTs between other developing countries in relation to the same origin country of investment. Thus, tax treaty bargaining does not take place in isolation, but depends on country's incentives to compete for mobile capital, reiterating the previous insights by Barthel and Neumayer (2012) and Arel-Bundock (2017).

Furthermore, where Barthel and Neumayer (2012) study the binary outcome of tax treaty formation, our analysis extends their spatial framework to the withholding tax rates negotiated by the two signatory countries. Similarly, we complement Arel-Bundock (2017) presenting new evidence on the relationship between concluded withholding tax rates and average withholding tax rates from competing countries' DTTs, employing a more robust empirical framework.

The remainder of the paper is structured as follows: Sect. 2 presents and motivates our hypothesis; Sect. 3 discusses our sample and empirical methodology; we

⁴ Other studies employing a network approach to analyse the effects of tax treaties include van't Riet et al. (2018) and Hong (2018).

present our main results in Sect. 4 and several robustness tests in Sect. 5. The paper concludes discussing its policy implications in Sect. 6.

2 Motivation

2.1 Stylised facts

DTTs remain an important policy tool to promote international economic activity with the total number of DTTs growing rapidly over the last 40 years. Contributing to this trend, DTTs concluded by and with developing countries have become increasingly more prevalent from the mid 1990s onwards. Focusing on withholding tax rates, Heckemeyer et al. (2024) show that developing countries conclude higher withholding tax rates with other developing countries than with developed countries. In their sample, tax treaties concluded between low-income countries reduce the withholding tax on participation dividends rate to 10%, on average, in contrast to an average tax rate of only 3.33% in tax treaties between low- and high-income countries, with similar patterns for withholding tax rates on interest and royalties. This indicates that developing countries give up a larger part of their taxing rights when negotiating DTTs with developed countries. The associated risk of tax revenue loss (see also Janský and Šedivý 2019) warrants a thorough examination of the determinants of withholding tax rates concluded in DTTs between developed and developing countries.⁵

In Fig. 1, we demonstrate the average withholding tax rate across all DTTs in any given year, distinguishing between DTTs signed between two developed countries (left panel) and DTTs signed between a developed and a developing country (right panel). Figure 1 shows a clear divergence in the evolution of the average withholding tax rates between these two groups over time. The average withholding tax rate (across all types of withholding taxes) signed between two developed countries remained fairly constant, starting at around 8% in 1980 and showing a slight decline only in the last few years of the sample. In contrast, the average withholding tax rate signed under DTTs between a developed and a developing country remained at a relatively high level of around 11.5% during the 1980s and declined sharply starting from 1990, reaching around 9% in 2016.

Figure 2 presents a more detailed picture by distinguishing among the four distinct types of withholding tax rates. First, we note that the modest decrease in average withholding tax rate observed for DTTs signed among developed countries in the last few years of the sample seems to be driven by the withholding tax rates on participation dividends. A particular example would be DTTs among EU Member States that often remove withholding tax rates for qualifying companies altogether

⁵ We acknowledge that it would be also interesting to examine the determinants of withholding tax rates concluded in DTTs between developing countries and more generally, when the balance of power of the two contracting states is more symmetric. However, in Sect. 3, we elaborate why the empirical methodology is more suited to analyse DTTs concluded between developed and developing countries.

aligning with the EU Parent-Subsidiary Directive and their own domestic law. However, while such an external coordination device is absent for the large sample of developing countries, we observe an even stronger decline in their average withholding tax rates on participation dividends.

Next, while the average withholding tax rates on interest and royalties under DTTs signed between developed countries have been moderately increasing since 1990, we observe a steady decline under DTTs signed between developed and developing countries. Finally, while average withholding taxes on portfolio dividends remained nearly constant for the case of DTTs signed by two developed countries, we see a regular decline in the same rates in the case of DTTs signed between a developed and a developing country, even below the average rate seen in DTTs signed among developed countries.

We argue that these stylised facts are consistent with the notion of tax competition for mobile capital. In particular, Fig. 3 illustrates the development of average withholding tax rates seen previously in Fig. 1, but distinguishing both in destination country's development and its openness to capital flows. We continue to observe a slight decline in the average withholding tax rates concluded in DTTs between developed and developing countries that are *closed* to capital flows. However, the effect is much stronger for tax treaties signed by developing countries that are *open* to capital flows. While we test this difference formally in the empirical part of the paper, we note that this empirical trend is consistent with our hypothesis.

For a more theoretically driven argument, the next section discusses a simple model of tax competition in the spirit of Davies and Voget (2008).

2.2 Theoretical framework

Consider a setting where each firm locates in the region offering it the greatest equilibrium profits. Similar to the derivation of the Logit estimator (*see* Green 2007), the probability L_j that any given firm m locates in a destination country j among the set of possible foreign locations S is:

$$L_j = \frac{\exp[(1 - T_j)\tilde{\Pi}_j]}{\sum_{s=1}^S \exp[(1 - T_s)\tilde{\Pi}_s]}, \quad (1)$$

with $\tilde{\Pi}_j$ and $\tilde{\Pi}_s$ standing for the equilibrium gross profits in countries j and s respectively (*see* Davies and Voget 2008). Differentiating with respect to country's j effective average tax rate T_j and country's s effective average tax rate T_s yields:

$$\frac{\partial L_j}{\partial T_j} = (L_j - 1)L_j\tilde{\Pi}_j < 0 \quad (2)$$

$$\frac{\partial L_j}{\partial T_s} = L_jL_s\tilde{\Pi}_s > 0 \quad (3)$$

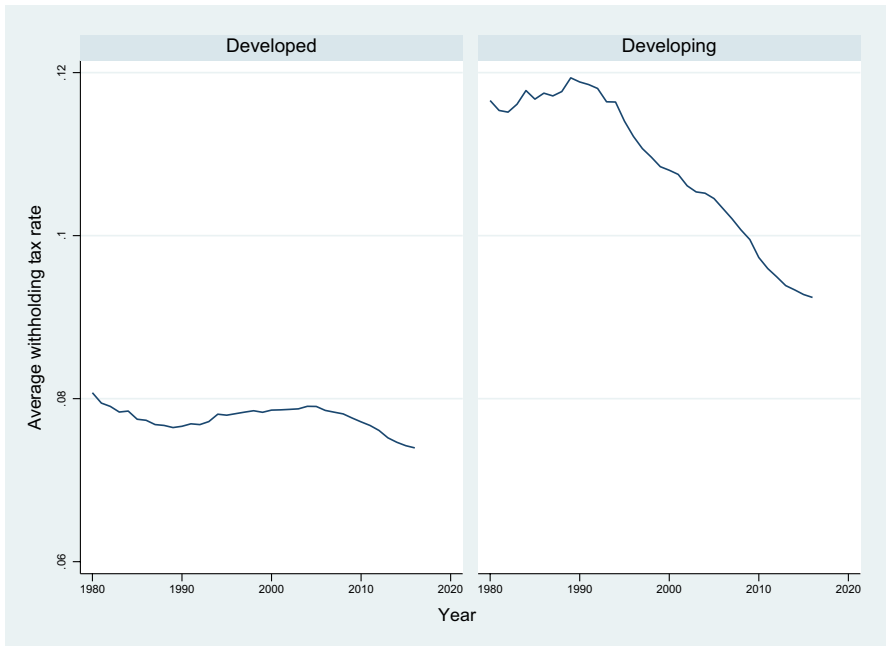


Fig. 1 Average withholding tax rate—by destination country development. The average withholding tax rate (across all types) under DTTs signed between two developed countries (left panel) and DTTs signed between a developed and a developing country (right panel). Source: authors' elaborations based on IBFD data (reported in Petkova et al. 2019), and UN M49 standard for statistical use for the distinction between developed and a developing countries

i.e. the probability of destination country j hosting firm m falls with its own effective average tax rate T_j and increases with the other country s effective average tax rate T_s . In Appendix A, we show formally how a reduction in destination country's j withholding tax rate decreases its effective average tax rate T_j , thereby establishing a direct link with our empirical setting.

Aggregating across the large number of firms implies that (in expected value) the equilibrium number of firms that destination country j hosts is L_j and its tax revenues are:

$$T_j L_j M \tilde{\Pi}_j, \quad (4)$$

where M is the total number of firms.

Governments simultaneously choose tax rates in order to maximise their own tax revenues. For country j , this yields an optimal value of tax:

$$T_j = (1 - L_j)^{-1} \tilde{\Pi}_j^{-1}, \quad (5)$$

where L_j depends on the tax rates of all countries S . From this, the best reaction function for country j with respect to the tax rate of country $s \neq j$ can be defined as:

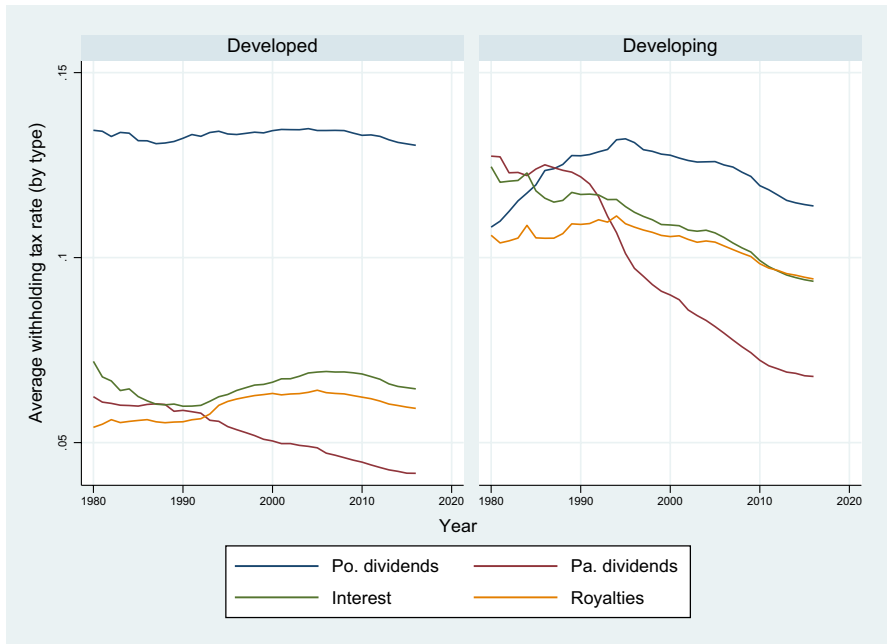


Fig. 2 Average withholding tax rate by type—by destination country development. The average withholding tax rate (by type) under DTTs signed between two developed countries (left panel) and DTTs signed between a developed and a developing country (right panel). *Po. dividends* and *Pa. dividends* refers to portfolio dividends, respectively participation dividends. Source: authors' elaborations based on IBFD data (reported in Petkova et al. 2019), and UN M49 standard for statistical use for the distinction between developed and a developing countries

$$\frac{\partial T_j}{\partial T_s} = \frac{L_j L_s \tilde{\Pi}_s}{(1 - L_j)^2 \tilde{\Pi}_j} > 0 \quad (6)$$

which indicates that tax rates are strategic complements. This predicts a positive effect of the spatially lagged withholding tax rates—as measured by the *PeerRate* variable—on the observed withholding tax rate in the empirical part of our analysis.

3 Data and empirical approach

We test our predictions empirically. In particular, we estimate a pooled cross-section across all years in our sample in the following baseline form:

$$WHT_{k,ij,t} = \alpha + \rho \omega_j \Omega_{is,t-n} + \beta X_{j,t} + \phi_j + \eta_t + \epsilon \quad (7)$$

where $WHT_{k,ij,t}$ is the negotiated withholding tax rate of type k between origin country i and destination country j in year of treaty conclusion t ; $\omega_j \Omega_{is,t-n}$ indicates our key explanatory variable, *PeerRate*—to be defined below; and $X_{j,t}$ is a vector of destination-country control variables capturing its bargaining position and incentives

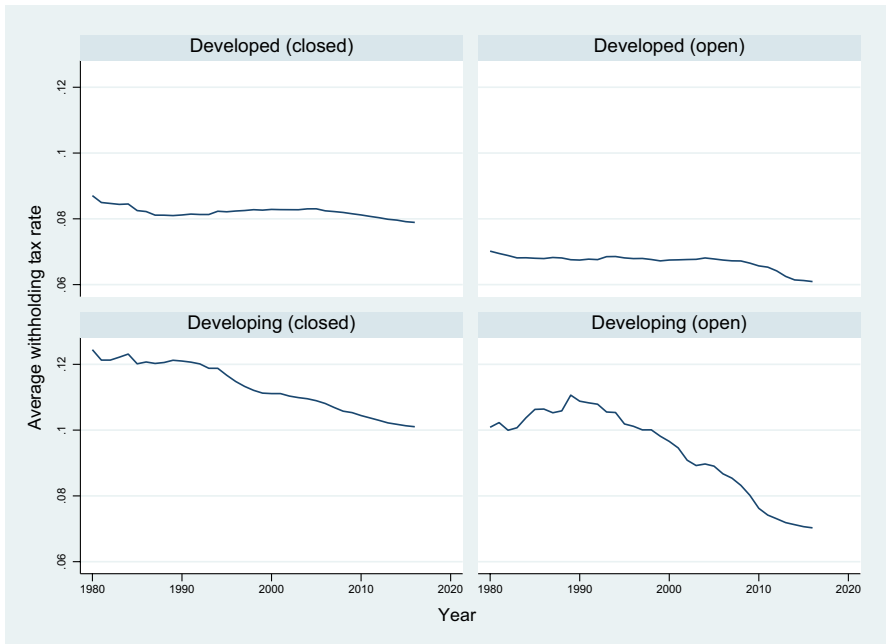


Fig. 3 Average withholding tax rate—by destination country development and openness to capital flows. The average withholding tax rate (across all types) under DTTs signed between a developed country and another developed country *closed* to capital flows (upper left panel); between a developed country and another developed country *open* to capital flows (upper right panel); between a developed and a developing country *closed* to capital flows (bottom left panel); and between a developed and a developing country *open* to capital flows (bottom right corner). Source: authors' elaborations based on IBFD data (reported in Petkova et al. 2019), and UN M49 standard for statistical use for the distinction between developed and a developing countries. Countries have been defined as *open* and *closed* using the the Gygli et al. (2019) KOF Globalisation Index de facto measure of economic globalisation, with a country defined as *open* if its index is above average for every year in the sample

to engage in tax competition, including GDP, GDP per capita, government expenditure and a dummy variable for openness to capital flows.⁶ In the baseline model specification, we employ destination-country, ϕ_j , and year, η_t , fixed effects, while in the alternative specifications, we resort to destination-year fixed effects, v_{jt} , and tax treaty fixed effects, θ_d . ϵ is the error term.

Withholding tax rates are generally available over a long time period once a DTT has been concluded. Nonetheless, they are typically constant over time. Thus, while a panel regression may appear desirable, the lack of within-treaty variation over time prevents us from estimating such a model (*see also* Chisik and Davies 2004).

⁶ In his meta-analysis, Heimberger (2021) shows that controlling for country size matters. In contrast, we do not find that controlling for partisan politics is statistically significant. One possible reason for this could be that our sample is centred around developing countries, while partisan politics are found to be statistically significant in studies centred around OECD and European countries (Osterloh and Debus 2012).

Instead, for each tax treaty we take a single observation in the year of treaty conclusion t and pool all available observations across time into a single cross-section. Consequently, we cluster standard errors by the destination country.

We construct the *PeerRate* variable by weighting the matrix of withholding tax rates between origin country i and all other potential destination countries $s \neq j$, i.e. Ω_{is} , by the weights given by (ω_j) .⁷ In the baseline specification of our model, all weights assume the value of unity—i.e. all other tax treaties are weighted equally—and the *PeerRate* takes the value of the arithmetic mean of peer countries' withholding tax rate. In the extensions, we use alternative definitions of the weight matrix ω_j to elicit heterogeneity stemming from geographical proximity, age and economic proximity.

While the *PeerRate* variable would normally suffer from endogeneity with respect to the dependent variable (i.e. with respect to itself), we leverage on the time dimension of treaty formation to mitigate these concerns. In particular, we resort to a space-time regression model and regress the observed withholding tax rate in the year of treaty conclusion t on the value of *PeerRate*, n years before. In this way, the *PeerRate* can only contain withholding tax rates from tax treaties that have been concluded before the observed withholding tax rate comes into existence. Throughout all of the main results, we estimate our model defining the time lag of the *PeerRate* variable with $n = 2$. In robustness tests, we replicate our analysis with different values of the time lag.

To illustrate this point, consider the following example in Table 1. The first DTT between Germany and the region of Southeast Asia is the tax treaty between Germany and Thailand signed in 1967.⁸ This treaty provides for a 15% withholding tax rate on royalties. Notice that up to this point, none of the destination countries observes a peer rate among its geographical peers. This is the mechanical consequence of the lack of treaties between Germany and countries in Southeast Asia prior to 1967.

With the 1967 DTT between Germany and Thailand, a single peer rate is established for all other destination countries to compete with, i.e. 15%, except for Thailand itself. Following the spatial econometric literature, we assume that a country cannot be “spatially” dependent on itself. Therefore, Thailand's withholding tax rate is excluded from Thailand's *PeerRate*.

The peer rate remains constant for all destination countries until 1972 when the DTT between Germany and Singapore comes into place. This treaty provides for a 0% withholding tax rate on royalties. Notice how the peer rate changes for each of the destination countries. From the perspective of Indonesia, Malaysia, Philippines

⁷ This is akin to a spatial weight matrix in the spatial econometric literature. However, we refrain from this terminology to make it clear that the matrix of withholding tax rates is not weighted in the baseline, neither in the “spatial” geographical sense, nor in some broadly defined economic proximity sense. In the extensions, when we weight the matrix of withholding tax rates by geographic proximity, age, and economic similarity measures, the *PeerRate* takes the classical form of a “spatial” weight matrix.

⁸ The example focuses on a specific region for illustrative purposes only. Our baseline model abstracts from geographical proximity.

and Vietnam, there are now two withholding tax rates to compete against in their geographical region: the 15% withholding tax rate levied by Thailand and the 0% withholding tax rate levied by Singapore, with the peer rate taking the average of these two values, i.e. 7.5%. Singapore continues to observe a single peer rate to compete against, i.e. the 15% withholding tax rate from the Germany-Thailand tax treaty. Finally, also Thailand observes a single peer rate to compete against, i.e. 0% negotiated between Germany and Singapore.

The year 1977 sees two new tax treaties coming into force: Germany-Indonesia and Germany-Malaysia, both providing for a withholding tax rate of 10%. Since both treaties come into force at the same point in time, taking the contemporary value of the peer rate would indeed suggest an effect of both tax treaties on each other. Hence, the independence of individual observations would be violated and OLS estimates would be biased.

However, in accordance with Anselin (2019), in the absence of temporal correlation, the spatial lag of the dependent variable from a different point in time than the observed dependent variable will be exogenous with respect to that dependent variable. To this end, our empirical approach regresses both observations on each destination countries' individual peer rate, 2 years before the year of treaty conclusion. Here, the peer rate takes the value of 7.5% in 1975 for both Indonesia and Malaysia. In this way, the OLS estimates will not be biased, and there is no need to resort to maximum likelihood in estimating the model.

We estimate our model for the four distinct treaty withholding tax rates on passive income—on portfolio dividends, participation dividends, interest and royalties—concluded between developed and developing countries between 1990 and 2016.⁹ Our time span of interest is motivated by the sharp decline in average treaty withholding tax rates starting from 1990 onwards. We hand collect these rates for more than 900 DTTs and amending protocols. We assume a 100%-owned subsidiary for the rate on participation dividends and collect interest rates commonly applied on inter-company loans as well as rates applicable to patent royalties. This set of assumptions ensures that we analyse withholding tax rates that are indeed in a competitive relationship vis-à-vis each other. Data on tax treaties and treaty withholding tax rates come from the IBFD Tax Research Platform (see Petkova et al. 2019 for a more detailed explanation of the underlying dataset).

We define countries as developed or developing in accordance with the UN M49 standard.¹⁰ Geographical groupings at the sub-region and intermediate region level

⁹ Focusing on DTTs concluded between developed and developing countries, we implicitly assume that it is developing countries that compete for investment from developed countries. In particular, extending our empirical setting to DTTs concluded between developing countries would require further assumptions on the assignment of origin and destination country. While this assumption seems plausible in our setting, assuming origin and destination countries in a setting with more equal countries seems more challenging. Ideally, one would resort to bilateral FDI statistics, which are however scarce for a sufficiently large number of countries and long enough time span. On the other hand, including a treaty-country-pair twice, with each country acting as an origin and destination country, raises issues of independence of observations. Against this background, we prefer to restrict the scope of the analysis to withholding tax rates concluded between developed and developing countries.

¹⁰ See <https://unstats.un.org/unsd/methodology/m49/>.

Table 1 Network of double tax treaties and peer rates

Time period	Tax treaties in force			Peer rates	
.				Indonesia	.
:				Malaysia	.
				Philippines	.
				Singapore	.
				Thailand	.
1966				Vietnam	.
1967	1967	Germany–Thailand:	15%	Indonesia	15%
:				Malaysia	15%
				Philippines	15%
				Singapore	15%
				Thailand	.
1971				Vietnam	15%
1972	1967	Germany–Thailand:	15%	Indonesia	7.5%
:	1972	Germany–Singapore:	0%	Malaysia	7.5%
				Philippines	7.5%
				Singapore	15%
				Thailand	0%
1976				Vietnam	7.5%
1977	1967	Germany–Thailand:	15%	Indonesia	8.33%
:	1972	Germany–Singapore:	0%	Malaysia	8.33%
	1977	Germany–Indonesia:	10%	Philippines	8.75%
	1977	Germany–Malaysia:	10%	Singapore	11.67%
				Thailand	6.67%
1982				Vietnam	8.75%

Columns 1 and 2 present the evolution of the network of DTTs between Germany and selected destination countries in Southeast Asia. Withholding tax rates in Column 2 apply to royalty payments. Column 3 shows the peer rate for the region

come from the UN M49 standard too. All of our control variables, including GDP and GDP per capita as well as the indicator of contiguity between countries, come from CEPII (Head et al. 2010). For a measure of openness, we resort to the KOF Globalisation Index (Gygli et al. 2019). In particular, we use the *de facto* measure of economic globalisation and define a country as *open* if its index is above average for every year in the sample.

4 Main results

We first estimate our baseline model for the four distinct types of withholding tax rates separately. Tables 2, 3, 4 and 5 present the baseline results for withholding tax rates on portfolio dividends, participating dividends, interest and royalties.

For brevity, we present only the estimates of the main variable of interest—*PeerRate*—and refer to Appendix B for the estimation results of control variables.

For each of the withholding tax rates, we first estimate our baseline model with year fixed effects only in Column (1). Next, we add destination-country fixed effects in Column (2) and destination-year fixed effects in Column (3). As many countries only sign one single tax treaty in any given year, country-year fixed effects remove much of the variation in the dependent variable. On the other hand, country-year fixed effects allow us to isolate the variation in withholding tax rates signed under different DTTs by the same destination country in a given year, effectively controlling for any country-year specific covariates. This yields a much more robust model specification, though possibly missing some heterogeneity in the results. By changing the structure of fixed effects, we can systematically compare different model specifications.

Using year fixed effects in Column (1), we find a positive and highly significant relationship between the observed withholding tax rate and the lagged average withholding tax rate from *past* tax treaties signed by other developing countries in relation to the same origin country of investment for three out of the four types of withholding tax rates: i.e. on portfolio dividends, interest and royalties. The estimated elasticities vary between 0.25 in the case of royalties to 0.5 in the case of portfolio dividends. Even though positive, the *PeerRate* variable is not statistically significant in the case of participation dividends (Table 3).

Among control variables, bigger economies, which are expected to have more bargaining power, tend to negotiate higher withholding tax rates, as indicated by the positive and statistically significant coefficient on the log of GDP. Conversely, richer developing countries, as measured by the log of GDP per capita, tend to negotiate lower withholding tax rates on participation dividends, interest and royalties, but higher rates on portfolio dividends. Though not directly comparable to previous research, we find that countries with higher shares of young and old populations and lower shares of government expenditure negotiate higher withholding taxes. Finally, we find that *open* economies negotiate lower withholding tax rates on portfolio dividends, but higher withholding taxes on royalties.

The baseline results remain virtually unchanged after adding destination-country fixed effects in Column (2). We note the much higher R^2 , suggesting that this model specification captures a larger share of the variation. Also, we note that adding destination-country fixed effects renders virtually all destination-country control variables insignificant. This is consistent with covariates capturing variation across countries, rather than across time.

Column (3) presents our preferred model specification, employing destination-year fixed effects. We isolate the variation in withholding tax rates across DTTs signed by the same destination country in the same year with different partners having a different *PeerRate*. Our main results remain highly significant at the 1% level, except for withholding tax rates on participation dividends. Compared to Column (1) and Column (2), the estimated elasticity is slightly lower for portfolio dividends—down from 0.5 in Column (1) to 0.48 in Column (3)—and slightly higher for interest and royalties: 0.37 and 0.31 up from respectively 0.35 and 0.25.

Table 2 Portfolio dividends

	(1)	(2)	(3)	(4)
<i>PeerRate</i>	0.5035 ^a (0.0576)	0.4922 ^a (0.0821)	0.4782 ^a (0.0827)	0.4644 ^a (0.0586)
Destination Controls	X	X		
Observations	895	884	607	3382
R-squared	0.389	0.541	0.699	0.715
Year FE	X	X		
Destination FE		X		
Destination*Year FE			X	
Treaty FE				X

Dependent variable: withholding tax rate on Portfolio Dividends. Column 1 and Column 2 covariates reported in Table 13, respectively Table 14 in Appendix. Robust standard errors (Column 1); clustered by destination country (Column 2); clustered by year*destination country (Column 3); or clustered by double tax treaty (Column 4) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 3 Participation dividends

	(1)	(2)	(3)	(4)
<i>PeerRate</i>	0.1067 (0.0705)	0.0785 (0.0737)	0.1448 (0.0967)	0.1631 ^a (0.0414)
Destination Controls	X	X		
Observations	535	526	276	3382
R-squared	0.305	0.583	0.685	0.715
Year FE	X	X		
Destination FE		X		
Destination*Year FE			X	
Treaty FE				X

Dependent variable: withholding tax rate on Participation Dividends. Column 1 and Column 2 covariates reported in Table 13, respectively Table 14 in Appendix. Robust standard errors (Column 1); clustered by destination country (Column 2); clustered by year*destination country (Column 3); or clustered by double tax treaty (Column 4) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

We note the following two results emerging from the first set of estimates. First, we do not find any evidence for tax competition in withholding tax rates on participation dividends. This seems particularly surprising given the strong decline in the average withholding tax rate on participation dividends documented in Fig. 2. However, as will be evident from Table 6, the distinction between *open* and *closed* economies is particularly relevant in this regard, with *open* economies competing also in withholding tax rates on participation dividends.

Table 4 Interest

	(1)	(2)	(3)	(4)
<i>PeerRate</i>	0.3509 ^a (0.0664)	0.3375 ^a (0.0742)	0.3747 ^a (0.0792)	0.1942 ^a (0.0580)
Destination Controls	X	X		
Observations	900	889	609	3382
R-squared	0.375	0.643	0.758	0.715
Year FE	X	X		
Destination FE		X		
Destination*Year FE			X	
Treaty FE				X

Dependent variable: withholding tax rate on Interest. Column 1 and Column 2 covariates reported in Table 13, respectively Table 14 in Appendix. Robust standard errors (Column 1); clustered by destination country (Column 2); clustered by year*destination country (Column 3); or clustered by double tax treaty (Column 4) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 5 Royalties

	(1)	(2)	(3)	(4)
<i>PeerRate</i>	0.2510 ^a (0.0515)	0.2675 ^a (0.0625)	0.3063 ^a (0.0639)	0.1621 ^a (0.0532)
Destination Controls	X	X		
Observations	902	891	610	3382
R-squared	0.318	0.617	0.755	0.715
Year FE	X	X		
Destination FE		X		
Destination*Year FE			X	
Treaty FE				X

Dependent variable: withholding tax rate on Royalties. Column 1 and Column 2 covariates reported in Table 13, respectively Table 14 in Appendix. Robust standard errors (Column 1); clustered by destination country (Column 2); clustered by year*destination country (Column 3); or clustered by double tax treaty (Column 4) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Second, we note that the tax competition effect on portfolio dividends seems stronger than that on other types of withholding tax rates. We speculate that developing countries are more willing to compete on withholding taxes on portfolio dividends (especially in absence of a reduced rate on participation dividends), as this leaves them less prone to profit shifting strategies. Specifically, dividends are paid from after-tax profits, i.e. after any behavioural choices that may result in profit shifting. In contrast, interest and royalty payments are deductible from the taxable base, thus creating a potential channel of profit shifting. Therefore, a lower withholding

Table 6 Tax versus yardstick competition

		(1)	(2)	(3)	(4)
<i>Portfolio Dividends</i>	<i>PeerRateOpen</i>	0.7757 ^a (0.1049)	0.8413 ^a (0.1450)	0.7418 ^a (0.1502)	0.5961 ^a (0.0996)
	<i>PeerRateClosed</i>	0.3793 ^a (0.0626)	0.3404 ^a (0.0733)	0.3299 ^a (0.0918)	0.4068 ^a (0.0554)
	H_o : <i>PeerRateOpen</i> =				
	<i>PeerRateClosed</i>	0.0009	0.0021	0.0201	0.0315
	Observations	895	884	607	3382
	R-squared	0.399	0.553	0.705	0.723
<i>Participation Dividends</i>	<i>PeerRateOpen</i>	0.3581 ^a (0.1167)	0.3276 ^a (0.1199)	0.6156 ^b (0.2496)	0.5701 ^a (0.1170)
	<i>PeerRateClosed</i>	0.0607 (0.0754)	0.0350 (0.0736)	0.0213 (0.0894)	0.0876 ^c (0.0468)
	H_o : <i>PeerRateOpen</i> =				
	<i>PeerRateClosed</i>	0.0156	0.0297	0.0270	0.0001
	Observations	535	526	276	3382
	R-squared	0.312	0.588	0.698	0.723
<i>Interest</i>	<i>PeerRateOpen</i>	0.6227 ^a (0.1419)	0.6693 ^a (0.1667)	0.7825 ^a (0.2046)	0.4193 ^a (0.1046)
	<i>PeerRateClosed</i>	0.2439 ^a (0.0762)	0.2184 ^a (0.0623)	0.2198 ^a (0.0744)	0.1289 ^b (0.0564)
	H_o : <i>PeerRateOpen</i> =				
	<i>PeerRateClosed</i>	0.0145	0.0098	0.0104	0.0026
	Observations	900	889	609	3382
	R-squared	0.382	0.650	0.764	0.723
<i>Royalties</i>	<i>PeerRateOpen</i>	0.4388 ^a (0.1019)	0.4693 ^a (0.1225)	0.5349 ^a (0.1256)	0.5014 ^a (0.1036)
	<i>PeerRateClosed</i>	0.1925 ^a (0.0604)	0.2099 ^a (0.0681)	0.2280 ^a (0.0702)	0.0908 ^c (0.0508)
	H_o : <i>PeerRateOpen</i> =				
	<i>PeerRateClosed</i>	0.0331	0.0576	0.0340	0.0000
	Observations	902	891	610	3382
	R-squared	0.322	0.620	0.758	0.723
	Destination Controls	X	X		
	Year FE	X	X		
	Destination FE		X		
	Destination*Year FE			X	
	Treaty FE				X

Dependent variable: withholding tax rate on Portfolio Dividends, Participation Dividends, Interest, or Royalties. Robust standard errors (Column 1); clustered by destination country (Column 2); clustered by year*destination country (Column 3); or clustered by double tax treaty (Column 4) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

tax rate on either one of these payments can leave the destination country prone to increased profit shifting strategies. In this context, the more limited scope to compete on withholding tax rates on interest and royalties is akin to the argument in Arel-Bundock (2017), who argues that treaty shopping imposes a policy constraint on countries' (domestic) tax policy.

We complement the empirical evidence presented so far and take a different approach in Column (4). Specifically, we pool the four distinct types of withholding tax rates into a single sample. In doing so, our sample includes now four observations for each tax treaty. This allows us to extend the baseline model specification with a tax-treaty fixed effect. This fixed effect will control for any unobserved heterogeneity in tax treaty characteristics, including the presence, scope and strength of individual provisions, such as tax information exchange, as well as any other origin-year, destination-year and country-pair characteristics. We further adapt our model by including dummies for the types of withholding tax rates other than the observed one, and interaction terms between each of them and the *PeerRate* variable. Thus, we are able to see the differential effect compared to the observed type withholding tax rate.

As before, we present only the estimates on the *PeerRate* variable in Tables 2, 3, 4 and 5 and refer to Appendix Table 15 Column (2) for the full regression results. We note several interesting results. First, the dummy variables for participation dividends, interest and royalties are all negative and statistically significant at the 1% level indicating that these rates are systematically lower than the withholding tax rates on portfolio dividends. Second, the *PeerRate* variable remains highly significant and positive pointing to statistically robust tax competition in negotiated withholding tax rates on passive income. Additionally, the *PeerRate* variable is now statistically significant also in the case of participation dividends. Finally, as before, we observe a much stronger tax competition effect in portfolio dividends than in the other three types of withholding tax rates.

A key issue in empirically identifying corporate tax competition is that it cannot be ruled out that the observed strategic interactions result from yardstick competition. It is also possible that countries follow each other in setting tax policy not because of any kind of competition, but because of common intellectual trends. Positive correlations between tax rates in different countries may therefore not necessarily imply competition for mobile capital (Devereux et al. 2008; Azémar et al. 2020).

We follow the established approach and disentangle the tax competition effect among countries that are *open* and *closed* to capital flows. This approach rests on the notion that yardstick competition does not require any capital flows into the two competing countries: it is as likely to take place between two completely closed (but otherwise similar) countries as between two open economies (Devereux et al. 2008). We resort to the KOF *de facto* economic globalisation index (Gygli et al. 2019) and define a country as *open* if its index is above average for every year in the sample.¹¹

We report the results across the four different types of withholding tax rates in Table 6. Columns (1) to (4) correspond to the four model specifications previously

¹¹ For a similar approach, see Azémar et al. (2020).

presented in Tables 2, 3, 4 and 5. Consistent with the notion of corporate tax competition, we find a positive and highly significant correlation between withholding tax rates in different DTTs for each type of withholding tax rates.

In contrast to the previous estimates, we find also a highly significant corporate tax competition effect among *open* economies for the withholding tax rates on participation dividends. This result suggests that the lack of statistically significant results in Table 3 is due to lack of tax competition from *closed* economies. A particular reason for this may be that it is indeed open economies that are more willing to include a reduced withholding tax rate on participation dividends and compete fiercer for foreign investment. In the baseline specification, this effect appears washed out by closed economies agreeing to include a reduced rate not to compete, but by means of political agenda.

Even though the *PeerRate* variable turns out statistically significant also among *closed* economies, we reject the hypothesis that this effect is as strong as among *open* economies. Thus, while we cannot rule out some form of yardstick competition, our results are consistent with the notion of tax competition for mobile capital.¹² The estimated elasticities suggest a much stronger tax competition effect among *open* economies with elasticities ranging from 0.53 for withholding tax rates on royalties to 0.78 for withholding tax rates on interest payments.

How do our results compare to those in previous literature? Our results are close to those reported by Azémar et al. (2020): the authors find corporate tax rate elasticities of around 0.44 to 0.49. Our baseline estimates on portfolio dividends are within this range, whereas our estimates on interest and royalties are slightly lower.

On the other hand, our results appear on the lower bound compared to some of the earlier papers focusing on developed countries. Devereux et al. (2008) find a corporate income tax rate elasticity of 0.67. Crabbé (2013) estimates the tax competition effect to vary between 0.57 and 0.79. Crivelli et al. (2015) find an elasticity of around 1.09 among OECD countries and around 0.61 among developing countries. This may suggest that (historically) tax competition between developed countries over corporate income tax rate has been stronger. However, once we isolate only the tax competition effect among *open* economies, our estimates are much more in line with those reported in previous literature. Overall, our results strongly support the notion of tax competition in withholding tax rates concluded between developed and developing countries.

¹² Our results are strengthened by the meta-regression results in Heimberger (2021), who shows that using economic openness as the weighting scheme tends to dampen the reported corporate tax competition effects. Thus, we should rather expect to find smaller estimates when splitting our sample in *open* and *closed* economies. Our results show the opposite effect.

5 Robustness tests

We validate our baseline results via various robustness tests. First, we address the potential endogeneity due to sample selection in Table 7 Panel A. Second, we relax one of the assumptions underlying our empirical model and allow for endogeneity in the *PeerRate* variable. We resort to a spatial autoregressive model and estimate the long-run equilibrium elasticity in Table 7 Panel B. In Table 8, we extend the baseline model and control explicitly for indirect tax rates. This allows us to verify that treaty shopping opportunities, which constrain state autonomy, do not alter our findings. Finally, Tables 9, 10, 11 and 12 present a diversity of regression results using alternative definitions of the baseline weight matrix. This allows us both to corroborate our baseline results and investigate the extent to which geography, tax treaty age and economic proximity alter tax competition in withholding tax rates.

Since withholding tax rates signed under a DTT are observable only for those country-pairs that have signed a DTT, our sample is subject to sample selection. However, we verify that this restriction does not bias our results by estimating the Heckman sample selection model. We construct the first-stage sample by including all country-pairs with a tax treaty in its year of signature and for the remaining country pairs we take the observation for the final year in our sample, i.e. 2016. For country-pairs with multiple tax treaties, we keep only the most recent one. Further, we satisfy the exclusion restriction by including former colony status as a predictor of tax treaty formation in the first stage.¹³

We present the estimates of the *PeerRate* variable in Table 7 Panel A and refer to Appendix Table 16 for the first step estimates and Appendix Table 17 for the estimates disaggregated between *open* and *closed* economies. Our main results remain unaffected and we continue to find highly significant elasticities for withholding tax rates on portfolio dividends and interest. Moreover, we find an even stronger result for the case of withholding tax rates on portfolio dividends than in our baseline model. In the case of withholding tax rates on royalties, the estimated elasticity is a bit lower, but remains significant at the 5% level. Overall, we continue to find statistically significant support for tax competition on all types of withholding tax rates among *open* economies (Appendix Table 17).¹⁴

Next, we relax one of the assumptions underlying our empirical model and allow for endogeneity in the *PeerRate* variable. As explained in Sect. 3, in absence of temporal correlation, the “spatial” lag of the dependent variable from a different point in time will be exogenous with respect to that dependent variable (Anselin 2019).

¹³ We verify that former colony status does not influence withholding tax rates by including this variable as an additional regressor in our main model. Indeed, we find no statistically significant relationship with our dependent variable (results available upon request).

¹⁴ As a further robustness check to sample selection, we analyse more formally withholding tax rates as a predictor of treaty formation. Guo and Fraser (2010) argue that if the independent variable of interest is not a significant predictor of the dependent variable potentially subject to endogeneity, there is no sample selection bias. We estimate a Dynamic Correlated Random Effects (CRE) Probit model and find no statistically significant relationship between withholding tax rates and treaty formation. Results available upon request.

Table 7 Sample selection bias and endogeneity

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
(A) <i>Sample Selection Bias</i>	<i>PeerRate</i>	0.5516 ^a	−0.0935	0.3683 ^a	0.1758 ^b
		(0.1128)	(0.1336)	(0.1184)	(0.0771)
	Destination Controls	X	X	X	X
	Observations	3414	2992	3417	3419
	Selected	858	518	861	863
	Nonselected	2556	2474	2556	2556
(B) <i>SAR Model</i>	<i>PeerRate</i>	0.4148 ^a	0.4513 ^a	0.4825 ^a	0.4107 ^a
		(0.0453)	(0.0443)	(0.0482)	(0.0513)
	Observations	556	556	556	556
	R-squared	0.853	0.861	0.790	0.860
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Control variables not reported (available upon request). Robust standard errors (clustered by destination-year in Panel B) in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 8 Tax competition and treaty shopping

		(1)	(2)	(3)	(4)
(A) <i>Baseline Model</i>	<i>PeerRate</i>	0.7616 ^a	0.5205 ^a	0.5075 ^a	0.3752 ^a
		(0.1279)	(0.1641)	(0.1377)	(0.1044)
	Observations	288	140	287	288
	R-squared	0.681	0.653	0.731	0.759
(B) <i>Extended Model</i>	<i>PeerRate</i>	0.8870 ^a	0.5216 ^a	0.5174 ^a	0.3754 ^a
		(0.1132)	(0.1757)	(0.1378)	(0.1054)
	<i>IndirectTaxRate</i>	0.1131 ^a	0.0336	0.0104	0.0087
		(0.0406)	(0.0637)	(0.0210)	(0.0142)
	Observations	279	132	278	279
	R-squared	0.713	0.656	0.735	0.764
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Sample restricted to time period 2005–2016. Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

However, if countries were perfectly forward-looking in negotiating their DTTs, our estimates could potentially suffer from endogeneity of the *PeerRate* variable.

We address this problem by estimating a spatial autoregressive model (SAR) throughout the entire time dimension of our dataset (Table 7 Panel B). The *PeerRate* variable will contain both DTTs concluded prior to the observed tax treaty as well

Table 9 Weight matrix sensitivity

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>N</i> =3	<i>PeerRate</i>	0.4359 ^a (0.0803)	0.0699 (0.0890)	0.3493 ^a (0.0784)	0.2916 ^a (0.0603)
	Observations	599	276	601	602
	R-squared	0.694	0.682	0.756	0.758
<i>N</i> =4	<i>PeerRate</i>	0.3866 ^a (0.0790)	0.1076 (0.0688)	0.3752 ^a (0.0810)	0.2644 ^a (0.0582)
	Observations	594	272	596	597
	R-squared	0.686	0.695	0.763	0.757
<i>N</i> =5	<i>PeerRate</i>	0.3673 ^a (0.0764)	0.1103 ^c (0.0639)	0.3478 ^a (0.0792)	0.2591 ^a (0.0569)
	Observations	585	268	590	591
	R-squared	0.679	0.692	0.759	0.753
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 10 Dimensions of tax competition: regional effects

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
(A) <i>Sub-Region</i>	<i>PeerRate</i>	0.5471 ^a (0.1393)	0.4472 ^a (0.1639)	0.3169 ^a (0.1114)	0.2293 ^b (0.1043)
	<i>PeerRateRegion</i>	0.1342 (0.0972)	−0.0324 (0.0539)	0.2416 ^a (0.0697)	0.1540 ^b (0.0623)
	Observations	436	158	442	448
	R-squared	0.726	0.677	0.798	0.775
(B) <i>Int.Region</i>	<i>PeerRate</i>	0.4730 ^a (0.1331)	0.4257 ^b (0.2044)	0.2632 ^b (0.1211)	0.1587 (0.1123)
	<i>PeerRateRegion</i>	0.1720 ^c (0.0954)	−0.0671 (0.0717)	0.2920 ^a (0.0769)	0.1683 ^b (0.0652)
	Observations	389	120	392	398
	R-squared	0.744	0.686	0.798	0.773
(C) <i>Contiguity</i>	<i>PeerRate</i>	0.2885 ^b (0.1156)	0.1449 (0.3640)	0.2572 ^b (0.1187)	0.1768 (0.1100)
	<i>PeerRateNeighbours</i>	0.1040 (0.1276)	0.1764 (0.2263)	0.0195 (0.0628)	0.0523 (0.0919)
	Observations	264	52	262	265
	R-squared	0.741	0.691	0.797	0.779
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 11 Dimensions of tax competition: dynamic effects

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
(A) <i>W = Age-Weighted</i>	<i>PeerRate</i>	0.5444 ^a (0.0839)	0.2992 ^a (0.0865)	0.4621 ^a (0.0719)	0.3469 ^a (0.0683)
	Observations	606	276	608	609
	R-squared	0.714	0.701	0.766	0.756
(B) <i>W = MostRecent</i>	<i>PeerRate</i>	0.3374 ^a (0.0806)	0.0112 (0.0897)	0.2991 ^a (0.0806)	0.2764 ^a (0.0682)
	<i>PeerRateMostRecent</i>	0.1815 ^a (0.0502)	0.2486 ^a (0.0525)	0.1212 ^a (0.0453)	0.0563 (0.0365)
	Observations	606	276	608	609
	R-squared	0.711	0.719	0.761	0.756
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 12 Dimensions of tax competition: economic proximity

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>W = GDPLead</i>	<i>PeerRate</i>	0.3351 ^a (0.0763)	0.0496 (0.0717)	0.0362 (0.1065)	0.1349 ^b (0.0636)
	Observations	606	276	608	609
	R-squared	0.678	0.681	0.744	0.741
<i>W = GDPSimilarity</i>	<i>PeerRate</i>	0.3086 ^a (0.0644)	0.0556 (0.0716)	0.1883 ^a (0.0642)	0.1555 ^a (0.0479)
	Observations	593	276	595	596
	R-squared	0.687	0.681	0.750	0.744
<i>W = PopLead</i>	<i>PeerRate</i>	0.2468 ^a (0.0634)	0.0651 (0.0712)	0.1107 (0.1126)	0.0733 (0.0502)
	Observations	607	276	609	610
	R-squared	0.672	0.682	0.745	0.740
<i>W = PopSimilarity</i>	<i>PeerRate</i>	0.2931 ^a (0.0494)	0.1175 (0.0771)	0.2656 ^a (0.0551)	0.1743 ^a (0.0537)
	Observations	594	276	596	597
	R-squared	0.690	0.685	0.760	0.747
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

as after it. We adopt a sample of 556 directed country-pairs for which we observe all four withholding tax rates. Notice that we are unable to separate the effect of previously concluded DTTs. In particular, including a spatial lag of the dependent variable that takes into account only previously concluded agreements, creates the problem of missing explanatory variable for the first DTTs in our sample. Hence, our results can be interpreted as the long-run equilibrium elasticity in negotiated withholding tax rates.

In line with the spatial econometric literature, in order to account for the endogeneity of the spatial lag of the dependent variable, we estimate the model by maximum likelihood. The results point to tax competition in withholding tax rates that is significant at the 1% level for each of the withholding tax rate types. Moreover, the estimated effect is slightly higher than the baseline effect suggesting that the intertemporal effect estimates the lower bound.¹⁵ Thus, we are confident that our baseline results are not biased by potential endogeneity of the *PeerRate* variable.

Next, we extend our baseline model and control explicitly for indirect tax rates. Arel-Bundock (2017) shows that treaty shopping can reduce the potency of fiscal policy, put pressure on governments to change their domestic tax laws, and ultimately, constrain state autonomy. In light of this, it appears necessary to validate if the positive relationship in concluded withholding tax rates is indeed stemming from a tax competition effect or resulting from binding indirect tax rates. To this end, we extend the approach from Petkova et al. (2019) and augment our empirical model with the indirect treaty shopping rate on participation dividends, interest and royalties. Specifically, we extend the data underlying the network analysis in Petkova et al. (2019) until 2016 and calculate indirect treaty shopping rates for interest and royalty payments.

The final sample spans the time period 2005–2016, which is considerable shorter than in our baseline model. To validate that our results do not capture any systematic changes over time, we first estimate the baseline model for this reduced time period and report the results in Table 8 Panel A. Note that we find a considerably higher tax competition effect than in our baseline sample for all the four types of withholding tax rates, with the tax competition effect being now also statistically significant for participation dividends.

In Table 8 Panel B, we extend the baseline model specification with the *Indirect-TaxRate* variable.¹⁶ The tax competition effect remains virtually identical to Panel A. Furthermore, we find little evidence for the notion that treaty shopping affects governments' capacity to negotiate withholding tax rates on a bilateral basis. While this

¹⁵ This contradicts somehow with the results reported by Heimberger (2021). He shows that using OLS estimator yields bigger tax competition effects than using an instrumental variable estimator. He speculates that on average, IV estimators indeed address underlying endogeneity issues that arise when governments simultaneously set their tax policy, while OLS is unable to do so.

¹⁶ Note that in the case of portfolio dividends, the indirect tax rate is not measured accurately as we were not able to differentiate effectively between portfolio and participation dividends for the withholding tax rates imposed under domestic law. Nonetheless, we believe that this does not alter the results in a significant matter. Furthermore, the evidence for the other types of withholding tax rates in Table 8 strongly rejects the notion that indirect treaty rates affect our baseline results.

result is different from Arel-Bundock (2017), we note that our setting and empirical methodology are ultimately quite different, focusing on withholding tax rates concluded in DTTs. In contrast, Arel-Bundock (2017) focuses on the bilaterally applicable withholding tax rate, whether as a matter of domestic law or under a DTT. In this context, countries may respond to policy constraints from treaty shopping by adjusting their domestic withholding tax rates.

Next, we validate the sensitivity of our results with respect to the chosen time lag of the *PeerRate* variable. Remember that throughout the main analysis we set $n = 2$. In Table 9, we replicate the main results for values $n = 3$, $n = 4$ and $n = 5$.¹⁷ Our main results remain unaffected. The withholding tax rate elasticities on portfolio dividends, interest and royalties remain statistically significant for all values of n , though their magnitude tends to decrease slightly for higher values of n .

In Table 10, we change the definition of the spatial weight matrix ω_j to analyse the tax competition effect at a regional level.¹⁸ Specifically, we extend the baseline model with a withholding tax rate matrix weighted by countries' geographic proximity. We define countries as spatially connected if both countries lie in the same *sub-region* in Panel A, if both countries lie in the same *intermediate region* in Panel B and if they share a common border in Panel C.¹⁹ Our results suggest a stronger tax competition effect at the regional level.

In particular, we find that the "baseline" tax competition effect at the *global* level is marginally lower than in our reference model. However, there is an additional reaction to changes in destination country *PeerRate* both at the *sub-region* and at the *intermediate region* level. The regional effect is substantial and increases the tax competition effect by more than half for withholding tax rates on interest and royalties. In contrast, the regional effect is absent at the *contiguity* level, though we note the much smaller sample size.²⁰

Next, we elicit the dynamic effects of tax competition in withholding tax rates. In Table 11 Panel A, we allow the spillover effect from past treaties to vary with their age.²¹ We speculate that more recently concluded DTTs may be a more relevant benchmark to compete against than older ones. In contrast to Table 10, we do not extend the model with a second withholding tax rate matrix, but change the definition of the weight matrix ω_j itself.²² To this end, we adapt our empirical specification by weighting our *PeerRate* variable inversely by the number of years that each of the past treaties has been in force in the signature year of the observed tax treaty. We find higher elasticities for all types of withholding tax rates and note that the

¹⁷ See Appendix Table 19 for the results disaggregated between *open* and *closed* economies.

¹⁸ We refer to Appendix Table 20 for the results disaggregated between *open* and *closed* economies.

¹⁹ We follow the UN M49 standard for country's regional classification.

²⁰ We test also a model specification resembling our baseline model, but changing the uniform weights for inverse distance weights. These results are very similar to our baseline results.

²¹ We refer to Appendix Table 21 for the results disaggregated between *open* and *closed* economies.

²² In Table 10, we could extend our baseline model with a second matrix of withholding taxes because of the low correlation between the two matrices. In contrast, inversely age-weighted matrix is highly correlated with the uniformly weighed matrix precluding the inclusion of both matrices in the same regression model.

PeerRate variable is now statistically significant also in case of withholding tax rates on participation dividends. This suggests that countries do indeed react stronger to DTTs signed more recently by their peers.

Does this effect extend to the single most recently signed DTT? In Table 11 Panel B, we analyse the tax competition effect from the most recently signed DTT by extending the baseline model with a weighted withholding matrix giving the most recent DTT a weight of unity and all other DTTs a weight of zero. As in our baseline model, we find statistically significant results at the 1% level for the withholding tax rates on portfolio dividends, interest and royalties. However, the magnitude is noticeably lower than before. In addition, we find a rather strong effect from the most recent treaty in the case of portfolio dividends and interest. Moreover, we find a strong tax competition effect among withholding tax rates on participation dividends, suggesting a more dynamic response than with the other types of withholding taxes.

Finally, we investigate to what extent economic proximity strengthens or dampens tax competition. A priori, it could be hypothesised that economically closer countries compete more fiercely against each other as they are more likely to compete for the same kind of foreign investment. On the other hand, Heimberger (2021) shows that using economic openness as the weighting scheme tends to dampen the reported corporate tax competition effects. To this end, we test four alternative definitions of the weighting matrix—defined as in Cassette and Paty (2008)—each capturing a different measure of economic proximity: *GDPLead*, *GDPSimilarity*, *PopulationLead* and *PopulationSimilarity*. Table 12 reports the results.²³

Using weights proportional to peer countries' GDP and population (i.e. *GDPLead* and *PopulationLead*), we find little evidence for tax competition, limited to withholding taxes on portfolio dividends and royalties. Put differently, there is little evidence to suggest that countries compete fiercer against economic leaders than against economically smaller countries. On the other hand, using weights proportional to peer countries' GDP and population similarity, we find results consistent with our baseline model, but of somewhat lower magnitude. This finding seems in line with Heimberger (2021), suggesting that economic proximity dampens the reported tax competition effects.

6 Conclusions

In our paper, we document a decline in average withholding tax rates in double tax treaties concluded between developed and developing countries after 1990 and empirically attribute this trend to tax competition. We find strong support for a positive relationship in withholding tax rates in tax treaties signed by developing countries in relation to the same origin country of investment. Depending on the type of the withholding tax rate, our results—validated across a wide range of robustness tests—suggest that the observed withholding tax rate will be between 0.31 pp. and

²³ See Appendix Table 22 for the results disaggregated between *open* and *closed* economies.

0.48 pp. lower for a 1 pp. lower average withholding tax rate from peer countries' past treaties. Moreover, we find a significantly stronger reaction among open economies, with elasticities ranging from 0.53 to 0.78.

We note two results. First, we find systematically stronger tax competition effect for the withholding tax rate on portfolio dividends than for the other types of withholding tax rates. We argue that this result is consistent with developing countries competing fiercer over a withholding tax rate that leaves them less exposed to tax avoidance and profit shifting strategies. Second, while our baseline results do not support the notion of tax competition in withholding tax rates on participation dividends, we show that when properly accounted for tax and yardstick competition, *open* economies compete equally strong over withholding tax rates on participation dividends.

Our paper contributes to various strands of research. First, we contribute to the literature on tax competition by showing that it extends beyond corporate income tax, as destination countries employ double tax treaties to bilaterally reduce withholding tax rates for foreign investors from a specific country of origin, leaving tax rates with respect to all other countries unaffected—a more targeted policy which we label bilateral tax competition. Moreover, while previous literature focuses primarily on tax competition between developed countries, we provide strong support for tax competition among developing countries for mobile capital from developed countries. We also contribute to the literature on tax treaty formation and more specifically, on tax treaty bargaining. In contrast to the previous literature, our paper does not view tax treaty negotiations as an exclusively bilateral outcome. Instead, we show that tax treaty bargaining does not take place in isolation, but depends on country's incentives to compete for mobile capital.

Our paper demonstrates the considerable scope for DTTs and their corresponding withholding tax rates as a tool of tax competition among developing countries. Future studies could further investigate the potential effect of the recent international agreement among the more than 140 member jurisdictions of the OECD/G20 Inclusive Framework on Base Erosion and Profit Shifting (IF) on the two-pillar solution to address the tax challenges arising from the digitalisation of the economy. In particular, the focus could be put on the effects of Pillar Two, i.e. the global effective minimum tax rate of 15%, on the development of the withholding tax rates. Overall, as Pillar Two aims to put a floor on tax competition and end the race-to-the-bottom in corporate tax rates, countries that wish to continue using tax policy to compete for inward investment might shift their attention to other areas of business taxation, including labour taxes, indirect taxes and withholding taxes at source—strengthening their importance and amplifying the need to understand their determinants. Finally, as tax treaties between developing countries will become more and more frequent, future research may look at the development of withholding tax rates concluded between countries with a more balanced power dynamic.

Appendix A

Devereux and Griffith (1999, 2003) define *EATR* for an international investment as:

$$T_j = \frac{R_j^* - R_j}{E(1 + \psi_j)p_j(1 + i)^{-1}} \quad (\text{A1})$$

where R_j^* is the pre-tax economic rent; R_j is the net present value of the economic rent generated by the perturbation of the subsidiary's capital stock; E is the expected exchange rate in period $t + 1$; ψ_j is the expected inflation in location j ; p represents the financial return; and i is the nominal interest rate in the presence of tax.

The net present value of the economic rent generated by the perturbation of the subsidiary's capital stock, R_j is defined as the sum of: (i) the rent attributable to the investment in the subsidiary financed by the retained earnings (R_j^{RE}); (ii) the additional cost of the parent raising external finance in the origin country h (F_h); and (iii) the additional cost of the subsidiary of raising finance from the parent (F_j). Thus, the economic rent earned from an investment by the subsidiary is given by:

$$R_j = R_j^{RE} + F_h + F_j \quad (\text{A2})$$

where only R_j^{RE} and F_j depend on cross-border taxes. From Eqs. (1) and (2) it follows that:

$$T_j = \frac{R_j^* - R_j^{RE} - F_h - F_j}{E(1 + \psi_j)p_j(1 + i)^{-1}} \quad (\text{A3})$$

In accordance with Devereux and Griffith (1999, 2003), we define finance from retained earnings as

$$\begin{aligned} R_j^{RE} = & -\gamma(1 - \sigma_{hj})(1 - A) \\ & + \frac{\gamma(1 - \sigma_{hj})}{(1 + \rho)} \{E(1 + \psi_j)(p_j + \delta)(1 - \tau_j) + E(1 + \psi_j)(1 - \delta)(1 - A_j)\} \end{aligned} \quad (\text{A4})$$

where γ is a term measuring the tax discrimination between new equity and distributions; A is the new present value of tax allowances per unit of investment; ρ is the ultimate shareholder's nominal discount rate; δ is the economic depreciation rate. Most importantly, σ_{hj} is the overall tax rate on dividend payments from the subsidiary located in country j to the parent resident in country h , which in turn, depends on the applicable withholding tax rate in the destination country j and the applicable method of double tax relief in the origin country of the parent h .

F_j depends on the source of subsidiary finance: retained earnings, new equity and debt. In particular, (with φ standing for tax depreciation),

$$F_j(\text{retained earnings}) = 0 \quad (\text{A5})$$

$$F_j(\text{new equity}) = \frac{\gamma\sigma_{hj}}{(1 + \rho)}(1 - \varphi_j\tau_j)[E - (1 + \rho)] \quad (\text{A6})$$

$$F_j(debt) = \frac{\gamma(1 - \varphi_j\tau_j)}{(1 + \rho)} \{ \sigma_{hj} [E(1 + i(1 - \tau_j)) - (1 + \rho)] - E\omega_{hj}i \} \quad (A7)$$

Similarly to σ_{hj} , $\omega_{hj}i$ is the overall tax rate on interest payment from the subsidiary to the parent that depends on the withholding tax rate in the country of the subsidiary j and methods of double tax relief in the origin country of the parent h .

Assuming a constant exchange rate E , it can be shown that R_j^{RE} and F_j both decrease with the applicable withholding tax rate of the destination country j . Formally,

$$\frac{\partial R_j^{RE}}{\partial \sigma_{hj}} = -\frac{\gamma}{(1 + \rho)} \{ E(1 + \psi_j)(p_j + \delta)(1 - \tau_j) + E(1 + \psi_j)(1 - \delta)(1 - A_j) \} < 0 \quad (A8)$$

$$\frac{\partial F_j(new\ equity)}{\partial \sigma_{hj}} = \frac{\gamma}{(1 + \rho)} (1 - \varphi_j\tau_j) [E - (1 + \rho)] < 0 \quad (A9)$$

$$\frac{\partial F_j(debt)}{\partial \omega_{hj}} = -\frac{\gamma Ei(1 - \varphi_j\tau_j)}{(1 + \rho)} < 0 \quad (A10)$$

Thus, a reduction in destination country's j withholding tax rate on dividends decreases its *EATR* on inbound foreign investments financed out of retained earnings and new equity.²⁴ A reduction in destination country's j withholding tax rate on interest payments decreases its *EATR* on inbound foreign investments financed by debt. Moreover, as long as the personal taxes in the residence country on capital gains are equal to or higher than the personal taxes in the residence country on interest income, a reduction in destination country's j withholding tax rate on dividends decreases also the *EATR* of a subsidiary financed by debt.²⁵ Hence, destination countries can compete for inbound foreign investment by reducing their withholding tax rates.

²⁴ Note that when a tax treaty between destination country j and origin country h is signed in period t , it becomes effective only in period $t + 1$. Hence, the first term of Eq. (8) is not relevant for its derivative. If we disregard the timing of the tax treaty, the theoretical predictions hold unambiguously for a greenfield investment not financed out of retained earnings.

²⁵ Formally,

$$\frac{\partial F_j(debt)}{\partial \sigma_{hj}} = \frac{\gamma(1 - \varphi_j\tau_j)}{(1 + \rho)} [E(1 + i(1 - \tau_j)) - (1 + \rho)] < 0, \text{ with} \quad (A11)$$

$$\rho = \frac{(1 - h^i)i}{1 - h^c}, \quad (A12)$$

where h^i is the residence country's personal tax rate on interest income; and h^c is the residence country's personal tax rate on capital gains.

Appendix B

See Tables 13, 14, 15, 16, 17, 18, 19, 20, 21, 22.

Table 13 Baseline model: control variables (Year FE)

	(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>PeerRate</i>	0.5035 ^a (0.0576)	0.1067 (0.0705)	0.3509 ^a (0.0664)	0.2510 ^a (0.0515)
<i>lnGDP</i>	0.0010 (0.0009)	0.0049 ^a (0.0010)	0.0075 ^a (0.0009)	0.0076 ^a (0.0008)
<i>lnGDPcap</i>	0.0049 ^c (0.0029)	−0.0081 ^a (0.0026)	−0.0059 ^b (0.0028)	−0.0094 ^a (0.0027)
<i>PopUrban</i>	−0.0004 ^a (0.0001)	−0.0000 (0.0001)	0.0002 (0.0001)	0.0003 ^b (0.0001)
<i>PopOld</i>	0.0026 ^a (0.0006)	−0.0005 (0.0005)	0.0031 ^a (0.0006)	0.0006 (0.0005)
<i>PopYoung</i>	0.0015 ^a (0.0003)	0.0003 (0.0003)	0.0021 ^a (0.0003)	0.0016 ^a (0.0003)
<i>GovExp</i>	−0.0003 (0.0003)	−0.0005 ^b (0.0002)	−0.0012 ^a (0.0003)	−0.0001 (0.0003)
<i>OpenDum</i>	−0.0269 ^a (0.0051)	0.0071 (0.0049)	0.0015 (0.0050)	0.0071 ^c (0.0041)
Observations	895	535	900	902
R-squared	0.389	0.305	0.375	0.318
Destination FE				
Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 14 Baseline model: control variables (destination & Year FE)

	(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>PeerRate</i>	0.4922 ^a (0.0821)	0.0785 (0.0737)	0.3375 ^a (0.0742)	0.2675 ^a (0.0625)
<i>lnGDP</i>	-0.0075 (0.0380)	-0.0374 ^b (0.0161)	0.0093 (0.0146)	-0.0164 (0.0200)
<i>lnGDPCap</i>	0.0018 (0.0298)	0.0238 ^c (0.0124)	-0.0103 (0.0127)	-0.0076 (0.0158)
<i>PopUrban</i>	0.0002 (0.0010)	0.0002 (0.0008)	0.0002 (0.0006)	0.0009 (0.0007)
<i>PopOld</i>	-0.0000 (0.0040)	-0.0032 (0.0028)	0.0001 (0.0034)	0.0001 (0.0043)
<i>PopYoung</i>	-0.0011 (0.0011)	-0.0014 (0.0009)	-0.0005 (0.0012)	-0.0024 ^c (0.0013)
<i>GovExp</i>	0.0005 (0.0005)	0.0003 (0.0005)	-0.0006 (0.0006)	-0.0003 (0.0005)
Observations	884	526	889	891
R-squared	0.541	0.583	0.643	0.617
Destination FE	X	X	X	X
Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 15 Pooled Model (Full specification)

	(1)	(2)
<i>PeerRate</i>	0.2781 ^a (0.0364)	0.4644 ^a (0.0586)
<i>PadDum</i>	−0.0524 ^a (0.0019)	−0.0197 ^a (0.0074)
<i>IntDum</i>	−0.0200 ^a (0.0016)	0.0110 (0.0077)
<i>RoyDum</i>	−0.0185 ^a (0.0017)	0.0157 ^b (0.0080)
<i>PeerRate*PadDum</i>		−0.3012 ^a (0.0645)
<i>PeerRate*IntDum</i>		−0.2702 ^a (0.0678)
<i>PeerRate*RoyDum</i>		−0.3022 ^a (0.0701)
Observations	3382	3,382
R-squared	0.711	0.715
Treaty FE	X	X

Dependent variable: withholding tax rates on portfolio dividends, participation dividends, interest and royalties. Robust standard errors clustered by double tax treaty in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 16 Sample Selection Bias (Heckman)—First Step

	(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>FormerColony</i>	1.2747 ^a (0.2379)	0.3692 (0.3272)	1.1817 ^a (0.2381)	1.2846 ^a (0.2389)
<i>PeerRate</i>	10.7299 ^a (0.9157)	32.4763 ^a (1.6524)	20.4357 ^a (1.5433)	16.1191 ^a (1.5514)
<i>lnGDP</i>	0.3048 ^a (0.0205)	0.2643 ^a (0.0249)	0.3053 ^a (0.0205)	0.2947 ^a (0.0201)
<i>lnGDPCap</i>	0.0371 (0.0535)	0.0507 (0.0683)	0.0430 (0.0539)	0.0487 (0.0541)
<i>PopUrban</i>	−0.0082 ^a (0.0020)	−0.0096 ^a (0.0026)	−0.0085 ^a (0.0021)	−0.0081 ^a (0.0020)
<i>PopOld</i>	−0.0353 ^a (0.0103)	−0.0146 (0.0128)	−0.0354 ^a (0.0104)	−0.0382 ^a (0.0103)
<i>PopYoung</i>	−0.0217 ^a (0.0057)	−0.0329 ^a (0.0074)	−0.0235 ^a (0.0059)	−0.0218 ^a (0.0058)
<i>Govexp</i>	−0.0168 ^a (0.0051)	−0.0142 ^b (0.0062)	−0.0170 ^a (0.0051)	−0.0179 ^a (0.0051)
<i>OpenDum</i>	0.1812 ^b (0.0751)	−0.0791 (0.1018)	0.1819 ^b (0.0785)	0.1794 ^b (0.0747)
Observations	3414	2992	3417	3419
Selected	858	518	861	863
Nonselected	2556	2474	2556	2556

Dependent variable: double tax treaty dummy taking the value of unity if a tax treaty is signed between origin and the destination state. Peer rate refers to the peer rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Estimated using the ML estimator. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 17 Sample Selection Bias and Endogeneity (Tax vs. Yardstick Competition Model)

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>Sample Selection Bias</i>	<i>PeerRateOpen</i>	0.9055 ^a (0.1292)	0.3913 ^b (0.1842)	0.6195 ^a (0.1866)	0.3727 ^a (0.1161)
	<i>PeerRateClosed</i>	0.4429 ^a (0.0929)	−0.0815 (0.1901)	0.2354 ^b (0.1095)	0.1076 (0.0841)
	Destination Controls	X	X	X	X
	Observations	3414	2992	3417	3419
	Selected	858	518	861	863
	Nonselected	2556	2474	2556	2556
<i>SAR Model</i>	<i>PeerRateOpen</i>	1.4842 ^a (0.1441)	1.3742 ^a (0.1444)	1.5920 ^a (0.1843)	1.3198 ^a (0.2217)
	<i>PeerRateClosed</i>	0.3606 ^a (0.0617)	0.4417 ^a (0.0646)	0.6377 ^a (0.1036)	0.5314 ^a (0.0870)
	Observations	556	556	556	556
	R-squared	0.614	0.829	0.070	0.776
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Control variables not reported (available upon request). Robust standard errors in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 18 Tax Competition and Treaty Shopping (Tax vs. Yardstick Competition Model)

		(1)	(2)	(3)	(4)
<i>Baseline Model</i>	<i>PeerRateOpen</i>	1.0444 ^a (0.1736)	0.8983 ^a (0.2487)	0.9155 ^a (0.2117)	0.4908 ^a (0.1563)
	<i>PeerRateClosed</i>	0.4938 ^a (0.1783)	0.3202 ^c (0.1767)	0.0999 (0.1309)	0.2246 ^b (0.1101)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.0290	0.0631	0.0014	0.1667
	Observations	288	140	287	288
	R-squared	0.693	0.669	0.745	0.761
<i>Extended Model</i>	<i>PeerRateOpen</i>	1.0328 ^a (0.1763)	0.8931 ^a (0.2469)	0.9129 ^a (0.2110)	0.4923 ^a (0.1564)
	<i>PeerRateClosed</i>	0.7179 ^a (0.1480)	0.2908 (0.1929)	0.1071 (0.1312)	0.2160 ^c (0.1093)
	<i>IndirectTaxRate</i>	0.1041 ^b (0.0424)	0.0472 (0.0605)	0.0085 (0.0221)	0.0069 (0.0143)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.1774	0.0584	0.0016	0.1491
	Observations	279	132	278	279
	R-squared	0.716	0.672	0.749	0.767
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Sample restricted to time period 2005–2016. Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 19 Weight Matrix Sensitivity (Tax vs. Yardstick Competition Model)

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>N</i> =3	<i>PeerRateOpen</i>	0.6783 ^a (0.1586)	0.5805 ^b (0.2441)	0.7889 ^a (0.2029)	0.5456 ^a (0.1205)
	<i>PeerRateClosed</i>	0.3049 ^a (0.0835)	−0.0277 (0.0828)	0.1931 ^a (0.0697)	0.2213 ^a (0.0654)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.0383	0.0200	0.0059	0.0189
	Observations	599	276	601	602
	R-squared	0.699	0.696	0.764	0.762
	<i>N</i> =4				
	<i>PeerRateOpen</i>	0.6560 ^a (0.1665)	0.5201 ^b (0.2418)	0.8198 ^a (0.2053)	0.5109 ^a (0.1205)
	<i>PeerRateClosed</i>	0.2548 ^a (0.0796)	0.0288 (0.0655)	0.2349 ^a (0.0785)	0.1999 ^a (0.0615)
	$H_o : \text{PeerRateOpen} =$				
<i>N</i> =4	<i>PeerRateClosed</i>	0.0307	0.0523	0.0083	0.0224
	Observations	594	272	596	597
	R-squared	0.692	0.706	0.770	0.760
	<i>N</i> =5				
	<i>PeerRateOpen</i>	0.6811 ^a (0.1734)	0.4852 ^b (0.2416)	0.7692 ^a (0.2122)	0.4910 ^a (0.1145)
	<i>PeerRateClosed</i>	0.2380 ^a (0.0755)	0.0415 (0.0605)	0.2113 ^a (0.0780)	0.1946 ^a (0.0613)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.0201	0.0776	0.0143	0.0235
	Observations	585	268	590	591
	R-squared	0.687	0.701	0.765	0.757
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 20 Regional Effects (Tax vs. Yardstick Competition Model)

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>Sub-Region</i>	<i>PeerRateOpen</i>	0.3470 (0.2955)	0.9147 ^a (0.2514)	0.4970 ^b (0.2256)	0.3624 ^b (0.1495)
	<i>PeerRateClosed</i>	0.5650 ^a (0.1438)	0.1987 (0.1627)	0.2382 ^c (0.1381)	0.1517 (0.1400)
	<i>PeerRateRegionOpen</i>	0.5143 ^a (0.1615)	−0.0267 (0.0712)	0.4657 ^a (0.1204)	0.1346 ^c (0.0795)
	<i>PeerRateRegionClosed</i>	−0.0854 (0.0917)	−0.0143 (0.0590)	0.1128 ^c (0.0670)	0.1671 ^b (0.0820)
	<i>H_o : Open = Closed</i>	0.0950	0.0204	0.0036	0.3014
	Observations	436	158	442	448
	R-squared	0.744	0.696	0.809	0.776
	<i>Int.Region PeerRateOpen</i>	0.5845 ^c (0.3173)	0.7333 ^b (0.3056)	0.4388 ^c (0.2233)	0.3444 ^b (0.1509)
	<i>PeerRateClosed</i>	0.3189 ^a (0.1080)	0.2210 (0.2161)	0.1187 (0.1407)	0.0145 (0.1579)
	<i>PeerRateRegionOpen</i>	0.3445 ^c (0.1848)	−0.1189 (0.0757)	0.4754 ^a (0.1241)	0.1436 ^c (0.0862)
<i>Int.Region</i>	<i>PeerRateRegionClosed</i>	0.0396 (0.0907)	0.0048 (0.1064)	0.1597 ^b (0.0805)	0.1874 ^b (0.0887)
	<i>H_o : Open = Closed</i>	0.0138	0.2706	0.0030	0.1053
	Observations	389	120	392	398
	R-squared	0.759	0.693	0.809	0.776
	<i>Contiguity PeerRateOpen</i>	0.4168 (0.3761)	—	0.5738 (0.3494)	0.3992 ^b (0.1720)
	<i>PeerRateClosed</i>	0.2415 ^b (0.1042)	—	0.1714 (0.1110)	0.1001 (0.1355)
	<i>PeerRateNeighboursOpen</i>	0.4869 (0.3587)	—	0.2989 (0.3250)	0.0599 (0.0781)
	<i>PeerRateNeighboursClosed</i>	0.0311 (0.1141)	—	0.0148 (0.0606)	0.0625 (0.1226)
	<i>H_o : Open = Closed</i>	0.0704	—	0.1437	0.2185
	Observations	264	—	262	265
<i>Contiguity</i>	R-squared	0.752	—	0.801	0.781
	Destination*Year FE	X	—	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 21 (Tax vs. Yardstick Competition Model)

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>W = AgeWeighted</i>	<i>PeerRateOpen</i>	0.8692 ^a (0.1309)	0.7379 ^a (0.1581)	0.9327 ^a (0.1346)	0.6004 ^a (0.1378)
	<i>PeerRateClosed</i>	0.3381 ^a (0.0928)	0.1737 ^b (0.0775)	0.2752 ^a (0.0643)	0.2584 ^a (0.0743)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.0011	0.0018	0.0000	0.0300
	Observations	606	276	608	609
	R-squared	0.725	0.713	0.776	0.760
	<i>W = MostRecent</i>				
	<i>PeerRateOpen</i>	0.4626 ^a (0.1541)	0.2475 (0.2591)	0.6641 ^a (0.2028)	0.4437 ^a (0.1206)
	<i>PeerRateClosed</i>	0.2655 ^a (0.0893)	-0.0655 (0.0917)	0.1654 ^b (0.0808)	0.2232 ^a (0.0785)
	<i>PeerRateMostRecentOpen</i>	0.3388 ^a (0.0932)	0.3982 ^a (0.1409)	0.1620 ^c (0.0840)	0.1304 ^b (0.0532)
	<i>PeerRateMostRecentClosed</i>	0.0807 ^c (0.0478)	0.1994 ^a (0.0550)	0.0932 ^c (0.0518)	0.0125 (0.0471)
	$H_o : \text{Open} =$				
	<i>Closed</i>	0.0073	0.0078	0.0087	0.0231
	Observations	606	276	608	609
	R-squared	0.723	0.731	0.767	0.761
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Table 22 Economic Proximity (Tax vs. Yardstick Competition Model)

		(1) Port. Div	(2) Part. Div	(3) Interest	(4) Royalties
<i>W = GDPLead</i>	<i>PeerRateOpen</i>	0.4453 ^a (0.1316)	0.2689 (0.2193)	0.0698 (0.2526)	0.3389 ^b (0.1666)
	<i>PeerRateClosed</i>	0.2508 ^a (0.0899)	0.0090 (0.0725)	0.0208 (0.1082)	0.0798 (0.0681)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.2236	0.2627	0.8587	0.1513
	Observations	606	276	608	609
	R-squared	0.679	0.684	0.744	0.742
	<i>W = GDPSimilarity</i>				
	<i>PeerRateOpen</i>	0.4359 ^a (0.1327)	0.4776 ^b (0.2015)	0.4447 ^a (0.1397)	0.2494 ^a (0.0926)
	<i>PeerRateClosed</i>	0.2291 ^a (0.0618)	-0.0335 (0.0578)	0.0697 (0.0687)	0.1155 ^b (0.0550)
	$H_o : \text{PeerRateOpen} =$				
<i>W = PopLead</i>	<i>PeerRateClosed</i>	0.1592	0.0163	0.0169	0.2154
	Observations	593	276	595	596
	R-squared	0.690	0.696	0.756	0.745
	<i>W = PopSimilarity</i>				
	<i>PeerRateOpen</i>	0.3593 ^a (0.1077)	0.2688 (0.1846)	0.1698 (0.2550)	0.1523 (0.1217)
	<i>PeerRateClosed</i>	0.1676 ^b (0.0747)	0.0137 (0.0743)	0.0837 (0.1220)	0.0483 (0.0537)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.1450	0.2024	0.7610	0.4353
	Observations	607	276	609	610
	R-squared	0.674	0.685	0.745	0.740
<i>W = PopSimilarity</i>	<i>PeerRateOpen</i>	0.4768 ^a (0.0872)	0.5388 ^a (0.1713)	0.4867 ^a (0.1141)	0.2335 ^a (0.0889)
	<i>PeerRateClosed</i>	0.1745 ^a (0.0539)	-0.0028 (0.0629)	0.1166 ^b (0.0483)	0.1356 ^b (0.0666)
	$H_o : \text{PeerRateOpen} =$				
	<i>PeerRateClosed</i>	0.0036	0.0037	0.0031	0.3791
	Observations	594	276	596	597
	R-squared	0.697	0.705	0.768	0.748
	Destination*Year FE	X	X	X	X

Dependent variable: withholding tax rate on Portfolio Dividends (Column 1); Participation Dividends (Column 2); Interest (Column 3); and Royalties (Column 4). Robust standard errors clustered by destination-year in parentheses. ^a, ^b and ^c denote significance at the 1, 5 and 10 percent confidence level

Acknowledgements We would like to thank Ron Davies, Emmanuel Saez, Alfons Weichenrieder, as well as the participants of the EGIT Annual Meeting, the WU DIBT research seminar, the Oxford University Centre for Business Taxation, the CPB Workshop on “The End of Tax Havens?”, the Berlin-Vallendar conference and the Annual Mannheim Taxation conference for helpful comments. Assistance with the data from Daniel Kubelka is highly appreciated. We gratefully acknowledge financial support from the Austrian Science Fund (FWF) Grant W1235-G16, the generous OeNB Jubiläumsfonds Grant 17825 and limited funding from the Università del Piemonte Orientale.

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